# nanospice

A SPICE circuit simulator in 1000 lines of Rust

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**Abstract**

nanospice is a circuit simulator with the classic Berkeley SPICE algorithm set: modified nodal analysis, Newton-Raphson with junction limiting and two operating point fallbacks, sparse LU factorization, trapezoidal transient integration with local truncation error timestep control, and small-signal AC analysis. The entire solver lives in one Rust source file of at most 1000 nonblank, noncomment lines, enforced by a test, with no dependencies outside the standard library. This report walks through the file in order, derives each algorithm, and explains the design decisions the budget forced. The bar under each section heading shows where that code section sits in the line budget.

## 1 The budget as a design tool

The rule: one file, `src/main.rs`, at most 1000 lines after removing blank lines and comments, standard library only. A test reads the file, counts, and fails the build above the limit. Tests, example netlists, and comments are free. The budget forces every feature to displace another, which makes the cost of each decision explicit, and it keeps the simulator readable in one sitting. That is the purpose of the repository.

This report follows the source file from top to bottom. Every section of the report corresponds to a section of the file, marked by the same name the code uses in its section comments, and carries a bar showing which slice of the budget it occupies. Table 1 is the overview.

| code section | lines | cumulative |
|---|---|---|
| constants, error and output plumbing | 23 | 23 |
| numbers: `Num` trait, complex, unit suffixes | 78 | 101 |
| circuit data model | 61 | 162 |
| parser | 275 | 437 |
| sparse LU and stamp helpers | 121 | 558 |
| device models | 64 | 622 |
| solver: assembly, Newton, operating point | 142 | 764 |
| output selection | 26 | 790 |
| analyses and main | 210 | 1000 |

Table 1: Line budget by code section, in file order.

The parser is the largest section: the numerics of a circuit simulator are compact, the input format is not. Everything the budget excluded is listed with reasons in section 14.

## 2 Background: modified nodal analysis

`theory only` 0 lines

Everything downstream assembles one linear system per Newton iteration, so the formulation comes first. Plain nodal analysis writes Kirchhoff's current law at every node except ground: the sum of currents leaving each node is zero. A conductance $g$ between nodes $p$ and $n$ carries the current $g(v_p - v_n)$, and collecting terms produces the node conductance matrix. The formulation breaks on ideal voltage sources: a voltage source has no admittance description, its current is whatever the circuit demands.

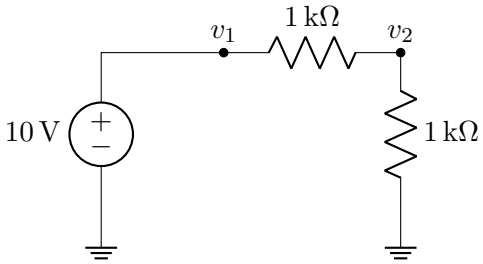
Modified nodal analysis (MNA) repairs this by promoting that current to an unknown. The vector of unknowns becomes

$$x = (v_1, \ldots, v_N,\ i_1, \ldots, i_B)^T,$$

with one branch current for every voltage-defined element: independent voltage sources, inductors (which are voltage-defined in DC, where they are shorts), and voltage-controlled voltage sources. Each such element contributes its current $\pm i_b$ to the KCL rows of its two terminal nodes and adds one constraint row of the form $v_p - v_n = E$.

A concrete example, the resistive divider used in the first integration test: a 10 V source into two 1 kΩ resistors, $g = 10^{-3}$ S,

$$\begin{pmatrix} g & -g & 1 \\ -g & 2g & 0 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ i \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 10 \end{pmatrix} \quad \Longrightarrow \quad v_1 = 10,\ \ v_2 = 5,\ \ i = -5\,\text{mA}.$$




Two properties matter later. First, the third diagonal entry is a structural zero: any LU factorization of an MNA matrix needs row pivoting. Second, $i$ comes out negative because the branch current is defined flowing into the positive terminal, the SPICE convention: a supply that delivers power reads negative.

Nonlinear devices fit the same frame through linearization, covered in section 9; reactive devices fit through companion models, also there. The result is that one matrix assembly routine serves DC, transient, and AC analysis.

## 3 Constants and plumbing

`constants, error and output plumbing` 23 lines, budget 0 to 23

The file opens with the numerical constants, all of them SPICE2 heritage:

| constant | value | meaning |
|---|---|---|
| `GMIN` | $10^{-12}$ S | conductance in parallel with every junction |
| `RELTOL` | $10^{-3}$ | relative convergence tolerance |
| `VNTOL` | $1\,\mu$V | absolute floor for voltage convergence |
| `ABSTOL` | 1 pA | absolute floor for current convergence |
| `TRTOL` | 7 | slack factor on the truncation error estimate |
| `VT` | 25.85 mV | thermal voltage $kT/q$ at 27 °C |
| `ITL_OP` | 100 | Newton iteration limit, operating point |
| `ITL_TRAN` | 10 | Newton iteration limit per transient step |

The two-part tolerance $\text{tol} = \tau + \text{RELTOL} \cdot |x|$ appears in both the Newton test and the timestep control. The relative part scales with signal amplitude; the absolute floor $\tau$ (VNTOL for voltages, ABSTOL for currents) keeps nodes that sit near zero from demanding impossible precision. TRTOL compensates for the conservative truncation error estimate; the value 7 is the SPICE2 default.

The remaining plumbing is a `die` helper that prints to stderr and exits, and an `out!` macro for stdout that exits quietly when the pipe closes. The macro exists because of a Unix detail: Rust ignores SIGPIPE, so a plain `println!` panics with a backtrace when the output is piped into `head` and the pipe closes.

# 4 Numbers

`numbers` 78 lines, budget 23 to 101



AC analysis needs complex arithmetic and the standard library does not provide it. Rather than duplicating the linear solver for `f64` and a complex type, everything numeric is generic over a three-method trait:

```
trait Num:
    Copy
    + std::ops::Add<Output = Self>
    + std::ops::Sub<Output = Self>
    + std::ops::Mul<Output = Self>
    + std::ops::Div<Output = Self>
{
    fn zero() -> Self;
    fn one() -> Self;
    fn mag(self) -> f64;
}
```

`mag` is the pivoting magnitude: absolute value for reals, modulus for complex. The complex type `Cx` implements the four operator traits in the obvious way; division uses the textbook formula

$$\frac{a + jb}{c + jd} = \frac{(ac + bd) + j(bc - ad)}{c^2 + d^2},$$

which is numerically adequate here because AC matrices of moderate size do not produce the extreme exponent spreads that would call for Smith's algorithm.

**Design decision: one generic solver instead of two copies**

The trait plus the operator implementations cost about 45 lines. Two copies of the sparse factorization would cost about 55 and could drift apart silently. The generic version also expresses the pivoting strategy once, through `mag`, instead of duplicating it.

The section closes with the SPICE number grammar: the longest prefix of a token that parses as a float, followed by a unit suffix from {`t`, `g`, `meg`, `k`, `m`, `u`, `n`, `p`, `f`}. The implementation shortens the token until `str::parse` succeeds, then looks the remainder up in a table ordered so that `meg` matches before `m`:

```
fn num(tok: &str) -> Option<f64> {
    let t = tok.trim();
    if !t.is_ascii() {
        return None;
    }
    let mut end = t.len();
    while end > 0 && t[..end].parse::<f64>().is_err() {
        end -= 1;
    }
    if end == 0 {
        return None;
    }
    let v: f64 = t[..end].parse().unwrap();
    let mult = SUFFIXES.iter().find(|(s, _)| t[end..].starts_with(s)).map_or(1.0,
    |(_, m)| *m);
    Some(v * mult)
}
```

The ASCII guard is a bug fix with a fuzz test behind it: Rust string slicing is byte-indexed and panics inside a multibyte character, so a netlist containing a literal micro sign must produce *bad number*, not a crash.

## 5 The circuit data model

`circuit data model` 61 lines, budget 101 to 162



A circuit is four vectors: devices, their names, node names, and the requested analyses, plus the branch count. Devices are one enum:

```
enum Dev {
    R { p: usize, n: usize, v: f64 },
    C { p: usize, n: usize, v: f64, ic: f64, m: f64 },
    L { p: usize, n: usize, v: f64, br: usize, ic: f64 },
    V { p: usize, n: usize, dc: f64, ac: f64, wave: Option<Wave>, br: usize },
    I { p: usize, n: usize, dc: f64, ac: f64, wave: Option<Wave> },
    D { p: usize, n: usize, is: f64, nf: f64 },
    M { d: usize, g: usize, s: usize, kp: f64, vt0: f64, lambda: f64, pmos: bool
    },
    E { p: usize, n: usize, cp: usize, cn: usize, k: f64, br: usize },
    G { p: usize, n: usize, cp: usize, cn: usize, k: f64 },
```

```
    Q { c: usize, b: usize, e: usize, is: f64, bf: f64, br: f64, pnp: bool },
}
```

**Design decision: an enum, not a device trait**

The alternative is a `Device` trait with a `stamp` method per implementation, which costs a trait definition, one `impl` block per device, and dispatch plumbing. The enum costs one declaration and a `match` arm per device at each use site, and the compiler checks exhaustively that no device is missing in any of them: assembly, output naming, breakpoint collection, state initialization. Three such omissions were caught at compile time while features were added. The device set is closed, so the extensibility a trait would buy is not needed.

Source waveforms are a second enum with the standard SPICE semantics. The damped sine, for $t \geq t_d$:

$$v(t) = v_o + v_a\, e^{-\theta(t-t_d)} \sin\bigl(2\pi f(t-t_d)\bigr),$$

and the trapezoidal pulse, piecewise linear through delay, rise, on, fall, off, optionally periodic. Initial conditions on C and L use `f64::NAN` as the *not specified* sentinel, which saves an `Option` wrapper and its pattern matching at every use site; NaN is unambiguous here because a NaN initial condition is meaningless anyway.

## 6 The parser

`parser` 275 lines, budget 162 to 437



The parser is the largest section. Preprocessing runs once per line: strip trailing comments after `;`, lowercase, replace parentheses and commas by spaces, then apply the line discipline. The first line of the file is a title, `*` starts a comment line, and `+` splices a continuation onto the previous line. Replacing the parentheses early means `sin(0 1 1k)` and `sin 0 1 1k` tokenize identically, and no grammar for balanced parentheses exists anywhere in the program.

Model cards are collected in a prepass over all lines so that `.model` may appear anywhere in the deck, before or after the devices that reference it. When a device line references a model, the model's parameter tokens are spliced *behind* the inline tokens of the line. Parameter lookup returns the first match, so instance parameters override model parameters by construction, with no explicit precedence logic:

```
let expand = |from: usize| -> Vec<&str> {
    let (mut inline, mut modt) = (Vec::new(), Vec::new());
    for t in toks.iter().skip(from) {
        match models.get(*t) {
            Some(mt) => modt.extend(mt.iter().map(|s| s.as_str())),
            None => inline.push(*t),
        }
    }
    inline.extend(modt);
    inline
};
```

Source specifications (`dc`, `ac`, `sin`, `pulse`, `pwl`) are parsed by a small keyword machine that greedily collects the numeric tokens after each keyword; missing trailing parameters default to zero, matching SPICE. The pulse is not a waveform of its own: the parser turns it into a piecewise linear point list, the same representation the `pwl` source uses directly, so evaluation is one interpolation routine and the breakpoint scan of section 11 reads the corner times straight from the points. The `.print` card is reconstructed from the parenthesis-stripped token stream: `v out` pairs back into `v(out)`, which works because every output column has exactly the form `kind(name)`.

Errors go through `die` with a named message: unknown device letters, unparseable numbers, missing tokens, unknown sweep sources. A fuzz test feeds seven classes of malformed netlist to the binary and requires a clean named error and no panic for each.

**Design decision: junction capacitances by desugaring**

Junction and gate capacitances (`cjo` on the diode, `cje` and `cjc` on the BJT, `cgs` and `cgd` on MOSFET and JFET) are capacitances across a node pair, which is exactly the `C` device. The parser therefore desugars them: a device line with capacitance parameters pushes internal `Dev::C` instances alongside the device (with the junction orientation flipped for the pnp). No assembly, state, timestep, or AC code knows they exist; transient integration, LTE control, and the $j\omega C$ stamps apply to them unchanged. Junctions carry the grading exponent `mj` (default 0.5), gate oxide capacitances are linear, and the `m=` parameter is available on the user-level `C` card as well. With the capacitances present, transistors switch in finite time and a three-stage ring oscillator has a defined period; an integration test measures it and requires it to be stable.

**Design decision: no subcircuits**

`.subckt` costs roughly a hundred lines of name spacing, port mapping, and flattening. Real decks are hierarchical, so this is the most limiting omission in practice. It was cut in favor of the transistor models.

# 7 Sparse LU

`sparse LU` 121 lines, budget 437 to 558



Gaussian elimination factors $PA = LU$ with row pivoting and then solves by substitution. Dense, this is 25 lines and $O(n^3)$, and it was the first implementation; it was replaced once the test suite pinned the behavior. A sparse package with fill-reducing ordering does not fit the budget. The following design, including the stamp helpers, takes 105 lines.

The matrix rows live in the `Sys` struct as vectors of (column, value) pairs kept sorted by column, and the stamp helpers write into them; the ground row and column, index 0, are silently dropped at insertion:

```
fn quad(&mut self, p: usize, n: usize, cp: usize, cn: usize, g: T) {
    self.add(p, cp, g);
    self.add(p, cn, T::zero() - g);
    self.add(n, cp, T::zero() - g);
    self.add(n, cn, g);
}
```

```
fn branch(&mut self, p: usize, n: usize, br: usize) {
    self.add(p, br, T::one());
    self.add(n, br, T::zero() - T::one());
    self.add(br, p, T::one());
    self.add(br, n, T::zero() - T::one());
}
```

**Design decision: a dummy ground row**

Ground is unknown 0 and the solver loops start at 1. The alternative is an *if node is not ground* test inside every stamp, roughly two dozen branches across ten devices. One guard inside `add` replaces all of them, and $x_0 = 0$ is pinned after the solve. The conductance quad above also covers the VCCS (rows at the output nodes, columns at the controlling nodes), and `branch` writes the current column and voltage row of V, L, and E; every stamp in the file reduces to these two shapes.

The third helper covers every nonlinear device in the file. A static device model is a set of current branches: a current $i$ flowing from node $p$ to node $n$ whose value depends on one or more controlling node-pair voltages. Linearized at the Newton iterate, such a branch stamps one conductance quad per control and one right-hand side pair. This is the contribution operator of Verilog-A, $I(p, n) <+ f(v)$, as a ten-line method:

```
// Verilog-A style contribution: current i flows p -> n, linearized at
// the (possibly limited) evaluation point and shifted to the raw
// iterate; ctl lists control pairs with d i / d (x[a] - x[b]).
fn contrib(&mut self, p: usize, n: usize, i: T, ctl: &[(usize, usize, T)], x: &[T
    ]) {
    let mut ieq = i;
    for &(a, b, g) in ctl {
        self.quad(p, n, a, b, g);
        ieq = ieq - g * (x[a] - x[b]);
    }
    self.rhs(p, T::zero() - ieq);
    self.rhs(n, ieq);
}
```

A dual primitive covers the voltage-defined elements: `vcontrib` writes the branch row $v_p - v_n - \sum_k g_k(x_{a_k} - x_{b_k}) = v$, which is the VCVS with one control, the inductor companion with its own branch current as the control (the pair $(br, 0)$), and the plain voltage source with none.

**Design decision: the whole device set on two primitives**

Every device arm in the assembly is now one or a few primitive calls: R, C, I, G, diode, MOSFET, and JFET are current contributions (`contrib`), V, L, and E are voltage contributions (`vcontrib`), and the BJT is three current contributions. The hand-written stamps disappeared: the refactor removed more lines than the two primitives cost, and the freed budget paid for the JFET and the PWL source. Hand-derived Jacobian stamps, like the $3 \times 3$ BJT matrix in section 9, become a verification artifact instead of code.

**Derivation: the first-entry invariant**

The factorization maintains: *before elimination step $k$, every row not yet chosen as a pivot contains only columns $\geq k$.*
Base case: column 0 is never stored, so before step 1 all rows hold columns $\geq 1$. Step: at step $k$ the active rows that contain column $k$ have it eliminated, and the elimination removes that entry from the row; rows without column $k$ are untouched. Afterwards every active row holds only columns $\geq k+1$. $\square$
Consequence: if an active row has an entry in column $k$ at all, that entry is the *first* element of its sorted vector. Pivot search over column $k$ reduces to comparing first elements, and the first column of a row is exactly the elimination step that will consume it, which makes rows bucketable by column.

The row update against the pivot row is a merge of two sorted sequences. Fill-in, the entries that appear where the original matrix had none, falls out of the merge automatically as the union of the two column sets:

```
// row a minus f times pivot row p, both sorted
fn merge<T: Num>(a: &[(usize, T)], p: &[(usize, T)], f: T) -> Vec<(usize, T)> {
    let mut o = Vec::with_capacity(a.len() + p.len());
    let (mut i, mut j) = (0, 0);
    while i < a.len() && j < p.len() {
        if a[i].0 == p[j].0 {
            o.push((a[i].0, a[i].1 - f * p[j].1));
            i += 1;
            j += 1;
        } else if a[i].0 < p[j].0 {
            o.push(a[i]);
            i += 1;
        } else {
            o.push((p[j].0, T::zero() - f * p[j].1));
            j += 1;
        }
    }
    o.extend_from_slice(&a[i..]);
    for &(c, v) in &p[j..] {
        o.push((c, T::zero() - f * v));
    }
    o
}
```

Rows never move. A bucket list per column holds the rows whose current first entry is that column: step $k$ takes bucket $k$, picks the largest `mag` as the pivot, eliminates the rest, and pushes each eliminated row into the bucket of its new first column; a permutation records which row became which $U$ row for the back substitution. MNA makes pivoting mandatory: the constraint row of every voltage source has a structural zero on the diagonal, visible in the divider example of section 2. If a bucket is empty the trailing matrix has an empty column and is singular; the factorization returns the failing column index and the caller maps it to a node or branch name. The most common user error, a node with no DC path to ground, produces *singular matrix at node b, no conduction path?* instead of a generic failure.

The elimination work is the sum of the lengths of the merged rows, so it is proportional to the fill-in; on ladder and tree topologies in netlist order there is none and the whole factorization is linear in $n$. The buckets are what make the pivot search match that bound. The first version of this solver scanned every remaining row per column instead, an $O(n^2)$ total, and shipped

that way until the benchmark in section 12 exposed it beyond $10^4$ unknowns; the bucket version costs eight more lines and is 47 times faster at 30000 nodes. The remaining omission is Markowitz ordering, so the worst case still degrades toward dense fill-in, which for the circuit sizes a 1000-line simulator realistically sees is the right cut.

# 8 Device models

`device models` 64 lines, budget 558 to 622



This section holds the semiconductor mathematics: the diode evaluation, the junction limiting shared by diode and BJT, and the MOSFET square law.

**Derivation: the graded junction capacitance**

The capacitor is charge based: the state variable is the charge, and the companion current over a step is $i = \frac{k}{h}(Q(v) - Q_{prev})$ with $k = 2$ for the trapezoidal rule and $k = 1$ for the damped restart steps, contributed with the conductance $\frac{k}{h}C(v)$. The graded junction has

$$C(v) = C_{j0}\left(1 - \frac{v}{V_J}\right)^{-m}, \qquad Q(v) = \frac{C_{j0}V_J}{1-m}\left(1 - \left(1 - \frac{v}{V_J}\right)^{1-m}\right),$$

with $V_J = 1\,\mathrm{V}$ fixed. The expression diverges at $v = V_J$, so above $F_C V_J$ with $F_C = 0.5$ the capacitance continues as its first-order Taylor expansion and the charge as its integral, keeping both $C^1$. Setting $m = 0$ recovers $Q = Cv$, the plain linear capacitor, so one evaluation function serves both. A test biases a junction at $v = -3\,\mathrm{V}$, where $(1 - v/V_J)^{-1/2} = \frac{1}{2}$ exactly, and checks that the AC corner frequency doubles.

**Diode.** Shockley equation with emission coefficient $n$ and the global $g_{min}$ in parallel:

$$i = I_S\big(e^{v/nV_T} - 1\big) + g_{min}v, \qquad g_d = \frac{I_S}{nV_T}e^{v/nV_T} + g_{min}.$$

The exponent is clamped at 200 as an overflow backstop; in normal operation the limiter below keeps it far smaller.

**Derivation: the critical voltage of pnjlim**

Newton diverges on the bare exponential: an early iterate proposing $v = 5\,\mathrm{V}$ puts $e^{v/V_T} \approx e^{193}$ into the Jacobian. The classic limiter damps the proposed junction voltage once it passes the critical voltage $v_{crit}$, defined as the point of maximum curvature of the $i(v)$ characteristic. With $i' = (I_S/V_T)e^{v/V_T} =: u$ and $i'' = u/V_T$, the curvature is

$$\kappa(v) = \frac{i''}{(1+i'^2)^{3/2}} = \frac{u}{V_T(1+u^2)^{3/2}}, \qquad \frac{d\kappa}{du} = \frac{1-2u^2}{V_T(1+u^2)^{5/2}} = 0 \implies u = \tfrac{1}{\sqrt{2}}.$$

Since $u$ grows monotonically in $v$, the maximum is unique, and substituting back:

$$\frac{I_S}{V_T}e^{v/V_T} = \frac{1}{\sqrt{2}} \quad\Longrightarrow\quad v_{crit} = V_T \ln\frac{V_T}{\sqrt{2}\,I_S}.$$

For $I_S = 10^{-14}\,\mathrm{A}$ this is about 0.72 V. Beyond $v_{crit}$, `pnjlim` replaces a proposed step by a logarithmic one, $v \leftarrow v_{old} + V_T \ln\big(1 + (v_{new} - v_{old})/V_T\big)$, which advances the junction a

controlled amount per iteration regardless of how far Newton overshoots.

```
fn pnjlim(vnew: f64, vold: f64, vt: f64, vcrit: f64) -> f64 {
    if vnew > vcrit && (vnew - vold).abs() > 2.0 * vt {
        if vold > 0.0 {
            let arg = 1.0 + (vnew - vold) / vt;
            if arg > 0.0 { vold + vt * arg.ln() } else { vcrit }
        } else {
            vt * (vnew / vt).ln()
        }
    } else {
        vnew
    }
}
```

A wrapper, `junction`, adds the bookkeeping every junction needs: computing $v_{crit}$, remembering the last limited voltage per device, and reporting whether limiting actually changed the proposal, which the Newton loop consumes (section 9). The diode uses one junction, the BJT two, which is why the limiter memory is two values per device.

**MOSFET, level 1.** The square law with channel-length modulation, $v_{ov} = v_{GS} - V_{T0}$:

$$i_D = \begin{cases} 0 & v_{ov} \le 0 \\ K_P \left( v_{ov} v_{DS} - \frac{1}{2} v_{DS}^2 \right) (1 + \lambda v_{DS}) & 0 < v_{DS} < v_{ov} \\ \frac{1}{2} K_P v_{ov}^2 \, (1 + \lambda v_{DS}) & v_{DS} \ge v_{ov} \end{cases}$$

with the small-signal parameters obtained by differentiation, $g_m = \partial i_D / \partial v_{GS}$ and $g_{ds} = \partial i_D / \partial v_{DS}$; in saturation $g_m = K_P v_{ov}(1 + \lambda v_{DS})$ and $g_{ds} = \frac{1}{2} K_P v_{ov}^2 \lambda$, so a nonzero $\lambda$ is what gives the transistor a finite output resistance.

**Derivation: the effective frame and the vanishing signs**

The model must serve four cases: nmos and pmos, each with either terminal acting as the source. Define the polarity $s = +1$ for nmos, $-1$ for pmos, and choose effective terminals $(d_e, s_e)$ so that $v_{DS}^e = s\,(v_{d_e} - v_{s_e}) \ge 0$; evaluate the square law in this frame. The physical current into $d_e$ is $I = s\, i_D$. Now take any derivative, for example with respect to the gate voltage:

$$\frac{\partial I}{\partial v_g} = s \cdot \frac{\partial i_D}{\partial v_{GS}^e} \cdot \frac{\partial v_{GS}^e}{\partial v_g} = s \cdot g_m \cdot s = g_m .$$

Every derivative picks up the polarity twice, once from the current and once from the voltage transformation, and $s^2 = 1$. The conductance stamps are therefore identical for all four cases; only the right-hand side term carries $s$. One sign variable and a conditional swap replace a four-way case split:

```
fn mos_op(dv: &Dev, x: &[f64]) -> (usize, usize, f64, f64, f64) {
    let Dev::M { d, g, s, kp, vt0, lambda, pmos } = dv else { unreachable!() };
    let sg = if *pmos { -1.0 } else { 1.0 };
    let (ed, es) = if sg * (x[*d] - x[*s]) >= 0.0 { (*d, *s) } else { (*s, *d) };
    let vgs = sg * (x[*g] - x[es]);
    let vds = sg * (x[ed] - x[es]);
    let (id, gm, gds) = mos_eval(vgs, vds, *kp, *vt0, *lambda);
```

```
    (ed, es, sg * id, gm, gds)
}
```

**JFET.** The level 1 JFET is the square law again: saturation current $\beta(v_{GS} - V_{TO})^2$ with a depletion threshold $V_{TO} < 0$, and the triode region matches the MOSFET equations with $K_P = 2\beta$. The parser maps the `J` device onto the MOSFET variant with $K_P = 2\beta$ and a default $V_{TO} = -2\,\mathrm{V}$; evaluation, stamps, and AC analysis come along unchanged, and the gate junction is not modeled. The device shares the MOSFET parser arm, so it costs a conditional in the parameter defaults and nothing anywhere else.

## 9 The solver

`solver: assembly, Newton, operating point` 142 lines, budget 622 to 764

**Companion linearization.** A nonlinear device current $i(v)$ enters the linear system as its tangent at the current Newton iterate $v^{(k)}$:

$$i \approx g\,v + I_{eq}, \qquad g = \left.\frac{\partial i}{\partial v}\right|_{v^{(k)}}, \qquad I_{eq} = i(v^{(k)}) - g\,v^{(k)},$$

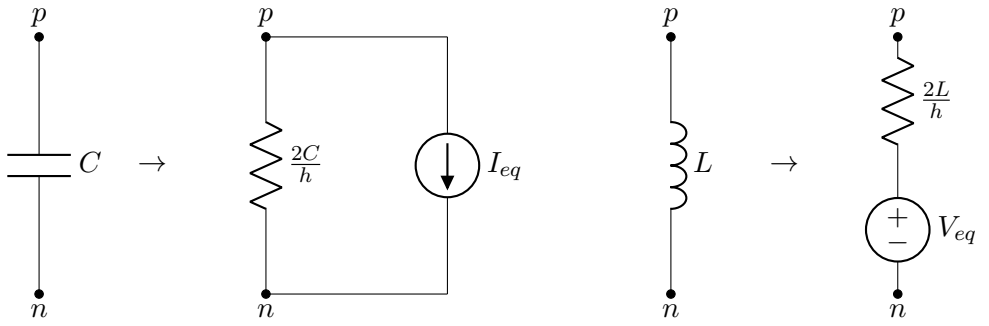
a conductance in parallel with a current source. Assembling and solving the resulting linear system is exactly one Newton-Raphson iteration on the full nonlinear network equations; the derivation is the multivariate Taylor expansion truncated after the linear term, applied device by device.

**Derivation: trapezoidal companion models**

In transient analysis the reactive devices become resistive companions. The trapezoidal rule applied to $i = C\,dv/dt$ over a step $h$:

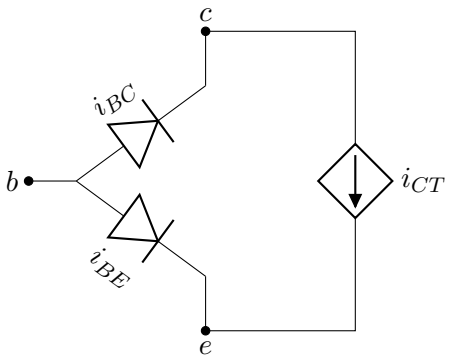
$$\frac{i_{n+1} + i_n}{2} = C\,\frac{v_{n+1} - v_n}{h} \implies i_{n+1} = \underbrace{\frac{2C}{h}}_{g_{eq}}\,v_{n+1} - \Big(\frac{2C}{h}\,v_n + i_n\Big),$$

a conductance $g_{eq}$ with a history current source: the capacitor is a resistor for the duration of one step. The inductor is the dual through the flux $\varphi = Li$ and $v = d\varphi/dt$, stamped into its branch row since its current is already an MNA unknown. Each reactive device keeps a two-entry state, its charge or flux and the dual rate ($i$ for the capacitor, $v$ for the inductor), and both duals advance by the same divided difference after every accepted step, one update rule for both element types.




Backward Euler would give $g_{eq} = C/h$ with a simpler history term and first-order accuracy; the trapezoidal rule is second order and A-stable with no numerical damping, which is why an LC tank simulated for twenty periods keeps its amplitude (a test enforces this). The price of zero damping is ringing at discontinuities, addressed by the breakpoint handling in section 11.

**The BJT.** The Ebers-Moll model in transport form is three current branches: the two base junctions and the transport source from collector to emitter.




With $e_f = e^{v_{BE}/V_T}$, $e_r = e^{v_{BC}/V_T}$:

$$i_{CT} = I_S(e_f - e_r), \qquad i_{BE} = \frac{I_S}{\beta_F}(e_f - 1), \qquad i_{BC} = \frac{I_S}{\beta_R}(e_r - 1),$$

$$I_C = i_{CT} - i_{BC}, \qquad I_B = i_{BE} + i_{BC}, \qquad I_E = -(I_C + I_B).$$

The transport form references one saturation current for the main transfer, which keeps forward and reverse operation consistent through a single $i_{CT}$. In the assembly, each branch is one contribution: the junctions with their own conductance, the transport source controlled by both junction voltages,

```
s.contrib(*b, *e, sg * (ibe + gpi * dbe), &[(*b, *e, gpi)], x);
s.contrib(*b, *c, sg * (ibc + gmu * dbc), &[(*b, *c, gmu)], x);
s.contrib(*c, *e, sg * (ict + gmf * dbe - gmr * dbc),
    &[(*b, *e, gmf), (*b, *c, -gmr)], x);
```

where `dbe` and `dbc` shift the linearization from the limited junction voltages back to the raw iterate. Expanding the three contributions by hand gives the classic $3 \times 3$ Jacobian stamp over

$(v_b, v_c, v_e)$, with $g_{mf} = \partial i_{CT}/\partial v_{BE}$, $g_{mr} = -\partial i_{CT}/\partial v_{BC}$, $g_\pi = \partial i_{BE}/\partial v_{BE}$, $g_\mu = \partial i_{BC}/\partial v_{BC}$:

$$\begin{pmatrix} g_{mf} - g_{mr} - g_\mu & g_{mr} + g_\mu & -g_{mf} \\ g_\pi + g_\mu & -g_\mu & -g_\pi \\ g_{mr} - g_{mf} - g_\pi & -g_{mr} & g_{mf} + g_\pi \end{pmatrix}$$

for the rows $I_C, I_B, I_E$. Rows and columns each sum to zero: shifting all node voltages together must change nothing, and KCL must close. In the contribution form this holds by construction, per branch. The pnp reuses the npn equations through the same sign argument as the MOSFET; both junctions go through `junction` with their own limiter memory. An integration test builds a common-emitter stage as npn and as pnp mirror image and requires exact symmetry of the solutions.

**Newton iteration and convergence.** Convergence is declared when every unknown moves less than its two-part tolerance between iterates, and at least two iterations have run so a fresh linearization has confirmed the point:

```
fn newton(ckt: &Circuit, x: &mut Vec<f64>, lim: &mut [[f64; 2]], mode: &Mode, gmin
    : f64, maxit: usize) -> Option<usize> {
    let nn = ckt.nodes.len();
    for it in 1..=maxit {
        let mut limited = false;
        let s = assemble(ckt, x, lim, mode, gmin, &mut limited);
        let xn =
            solve(s).unwrap_or_else(|k| die(&format!("singular matrix at {}, no
    conduction path?", unk_name(ckt, k))));
        let conv = (1..xn.len()).all(|i| (xn[i] - x[i]).abs() <= tolv(i, nn, xn[i
    ], x[i]));
        *x = xn;
        if conv && !limited && it > 1 {
            return Some(it);
        }
    }
    None
}
```

**Field note: limiting must deny convergence**

The `limited` flag records a bug found during development. With limiting alone, the following happened on a common-emitter amplifier: the bias divider put the base at 2.1 V while the emitter sat at ground, `pnjlim` clamped the base-emitter voltage seen by the stamps to 0.11 V, the transistor therefore conducted nothing, the node voltages stopped changing, and the delta-based convergence test declared success with the transistor off. The output was superficially plausible: the divider voltage was correct, but unloaded.
The repair is the classic one: an iteration in which a limiter changed a junction voltage is not allowed to converge. With the flag in place, the same circuit walks the junction up in logarithmic steps and reaches the correct bias, base at 2.05 V and 1.3 mA of collector current, in a dozen iterations. A convergence test on the solution vector alone cannot detect active limiting; the flag makes it visible.

**Continuation fallbacks.** When plain Newton fails, the operating point is found by continuation: replace the hard problem $F(x) = 0$ by a family $H(x, \lambda) = 0$ that is easy at $\lambda = 0$, coincides with $F$ at $\lambda = 1$, and is tracked by warm-started Newton solves along the path. Both

classic fallbacks are paths in the ($g_{min}$, source scale) plane, so the implementation is one loop over two lists of stages:

```
let gmin_path: Vec<(f64, f64)> = (2..=12).map(|k| (10f64.powi(-k), 1.0)).collect()
    ;
let src_path: Vec<(f64, f64)> = (1..=10).map(|k| (GMIN, k as f64 / 10.0)).collect
    ();
for path in [gmin_path, src_path] {
    // reset x and the limiter memory, then walk the stages warm started
    if path.iter().all(|&(g, fac)| newton(ckt, x, lim, &Mode::Dc { fac }, g,
    ITL_OP).is_some()) {
        return;
    }
}
```

Gmin stepping walks a junction conductance from $10^{-2}$ S down to $10^{-12}$ S decade by decade: at large $g$ the circuit is nearly linear and Newton converges reliably. Source stepping scales every source value from 0.1 to 1.0: the zero-source circuit has the trivial solution $x = 0$, and each increment moves the solution a small, trackable distance. Gmin stepping handles floating-junction trouble; source stepping handles feedback and bistable topologies, where gmin stepping can converge to the wrong branch. A cross-coupled NMOS latch in the test suite exercises the path.

## 10 Output selection

`output` — 26 lines, budget 764 to 790

Output columns are the node voltages in netlist order followed by the branch currents in device order, named `v(node)` and `i(device)`. A `.print` card filters this list by name; unknown names warn on stderr rather than fail, so a typo drops a column instead of aborting the run. The selection maps names to unknown indices once and every analysis row is then a gather over that index list.

## 11 Analyses

`analyses and main` — 210 lines, budget 790 to 1000



**Operating point and DC sweep.** `.op` is one call into the operating point solver. `.dc` sweeps a source value and re-solves at each point, warm-starting from the previous solution, a continuation along the sweep variable: on a smooth branch each solve takes two or three iterations. The sweep replaces the source specification; any transient waveform on the swept source is dropped.

**Transient timestep control.** The integrator adapts its step to the local truncation error (LTE) of the trapezoidal rule.

**Derivation: the Milne estimate and the cube-root rule**

The trapezoidal rule commits the local error $\varepsilon_{tr} = -\frac{h^3}{12}\,\dddot{x}(\xi)$ per step. The third derivative is estimated without extra solves by a predictor: a quadratic polynomial through the last three accepted points, extrapolated to the new time. With step history $d_1, d_2$ and new step $h$, Lagrange extrapolation gives

$$x_p = \ell_0\, x_n + \ell_1\, x_{n-1} + \ell_2\, x_{n-2},$$

$$\ell_0 = \frac{(h+d_1)(h+d_1+d_2)}{d_1(d_1+d_2)}, \qquad \ell_1 = -\frac{h(h+d_1+d_2)}{d_1 d_2}, \qquad \ell_2 = \frac{h(h+d_1)}{(d_1+d_2)\,d_2},$$

whose interpolation error at the new point is $\varepsilon_p = \frac{1}{6}\,\dddot{x}\; h(h+d_1)(h+d_1+d_2)$. Corrector and predictor err against the same true solution with errors proportional to the same $\dddot{x}$, so their difference measures it:

$$x_c - x_p = \varepsilon_p - \varepsilon_{tr} \quad\Longrightarrow\quad \varepsilon_{tr} \approx \frac{h^3/12}{h(h+d_1)(h+d_1+d_2)/6 \;+\; h^3/12}\,\big(x_c - x_p\big),$$

which for uniform steps ($d_1 = d_2 = h$) reduces to $\varepsilon_{tr} \approx (x_c - x_p)/13$, the classic factor. The error is compared per unknown against $\mathrm{TRTOL}\cdot\mathrm{tol}_i$; with the error scaling as $h^3$, the step that would exactly meet the tolerance satisfies $h' = h\,(1/r)^{1/3}$ where $r$ is the worst error ratio, and the controller applies it with a safety factor of 0.9, clamped to $[0.3, 2]\times h$ and capped at `tstep`. Rejected steps ($r > 1$) are retried smaller.

```
let etrap = hs * hs * hs / 12.0;
let fac = etrap / (hs * (hs + d1) * (hs + d1 + d2) / 6.0 + etrap);
let mut r: f64 = 0.0;
for i in 1..m {
    r = r.max(fac * (xn[i] - xp[i]).abs() / (TRTOL * tolv(i, nn, xn[i], x[i])));
}
let scale = 0.9 / r.max(1e-8).cbrt();
```

The predictor doubles as the Newton starting point, which is free and usually saves an iteration. Before three points of history exist, and after a nonconvergent step (cut by 8, the SPICE2 default), the iteration count controls the step instead. With `uic` the operating point is skipped entirely and the `ic=` values seed the reactive state, which is how an undriven LC tank can start oscillating at all: its operating point is identically zero.

**Field note: pulse edges inside a step**

The first RC test against the analytic step response failed at exactly the first output point. A pulse edge at $t \approx 0$ inside the first step $[0, h]$ is invisible to the method: the capacitor's history current at the interval start still belongs to the pre-edge circuit, so the trapezoidal update gives

$$v_1 = \frac{h/2\tau}{1 + h/2\tau} \approx 0.048 \quad \text{instead of} \quad 1 - e^{-h/\tau} \approx 0.095$$

for $h/\tau = 0.1$: a factor-two error at the first point, and no smaller $h$ fixes the ratio, because the history term is wrong by construction. The LTE controller cannot see it either, since the predictor has no history yet. The classic remedy is exact rather than adaptive: every point of every piecewise linear waveform (the pulse is parsed into its four corners) is registered as a breakpoint, steps are clamped to land on breakpoints exactly, and after crossing one the integrator restarts with a small step and cleared predictor history, because the waveform

derivative is discontinuous there.
The restart steps carry one more safeguard: while the predictor has no history, at startup and right after a breakpoint, the companion models switch from the trapezoidal rule to backward Euler. Backward Euler is first order but strongly damped, so the ringing the trapezoidal rule produces at discontinuities is absorbed in the two bootstrap steps, after which the undamped second-order rule resumes. This is the damped-restart scheme used by modern simulators, and it costs a boolean on the transient mode.

**Small-signal AC.** Write every waveform as a small phasor perturbation around the operating point, $x(t) = x_0 + \mathrm{Re}\{\hat{x}\, e^{j\omega t}\}$, insert into the network equations, and keep first-order terms:

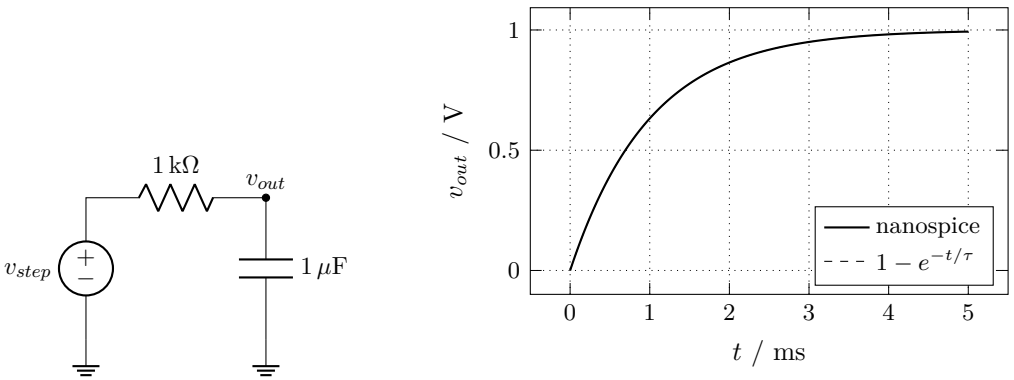
$$\big(G + j\omega C\big)\,\hat{x} = \hat{b},$$

where $G$ is the Jacobian of the resistive equations at $x_0$, $C$ collects the reactances, and $\hat{b}$ holds the `ac` source values. The implementation uses the following property: at a converged operating point the limiters pass voltages through unchanged, so $G$ *is exactly the matrix the last Newton iteration already assembles.* The AC path calls `assemble` once, maps the rows to complex, adds $j\omega C$ quads and the $-j\omega L$ branch entries, and hands the result to the same generic sparse LU. No device model contains any AC-specific code; adding a device to DC automatically adds it to AC. Frequencies sweep linearly or per decade ($f_k = f_1 \cdot 10^{k/n}$), and each unknown reports magnitude and phase.

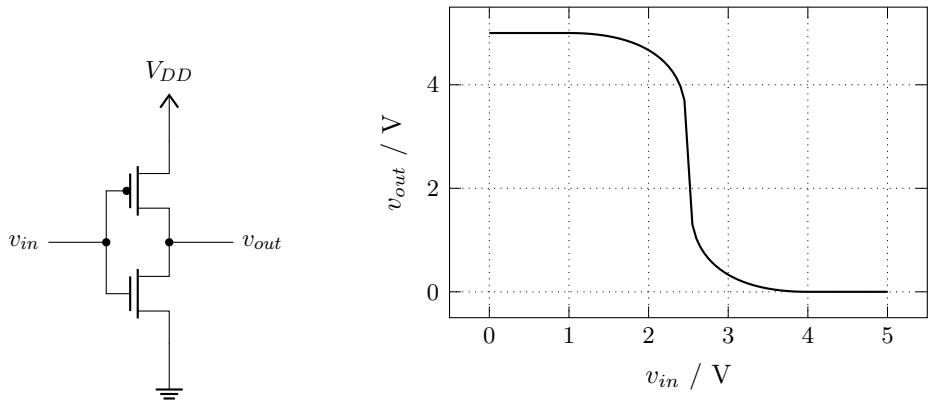
## 12 Reference circuits and results

Every plot in this section is simulator output: `report/data/gen.py` runs the release binary on the netlists shown and writes the data files the figures read. Analytic references are drawn dashed.

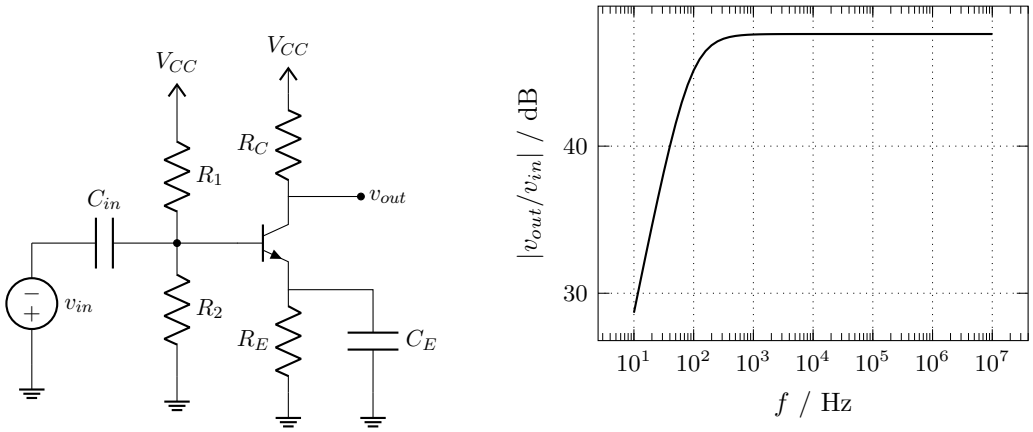
**RC step response.** The first transient test in the suite, plotted against its closed form:



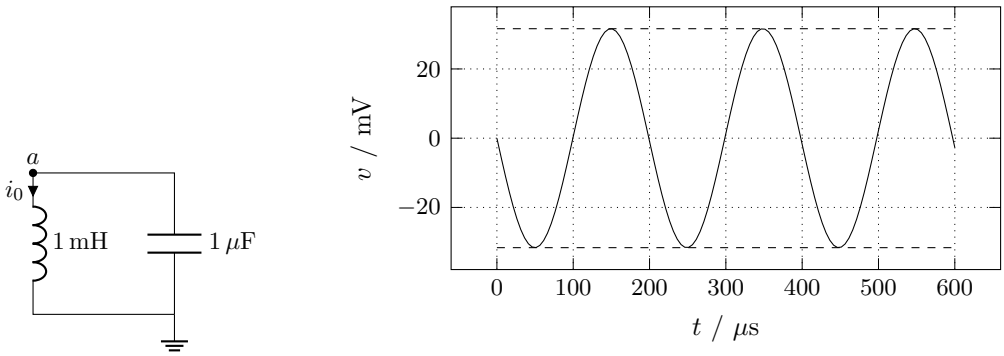

**CMOS inverter.** Two square-law transistors, swept with `.dc`; the finite transition slope is the channel-length modulation $\lambda = 0.05$:



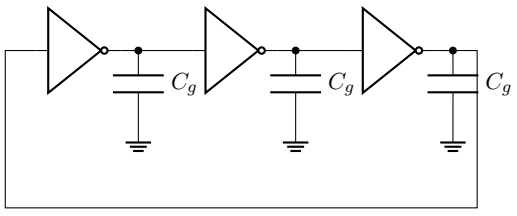

**Common-emitter amplifier.** Divider bias, emitter degeneration with a bypass capacitor, coupling capacitor at the input. Midband gain $g_m R_C \approx 240$ (47.6 dB); the low corner comes from the coupling and bypass capacitors:

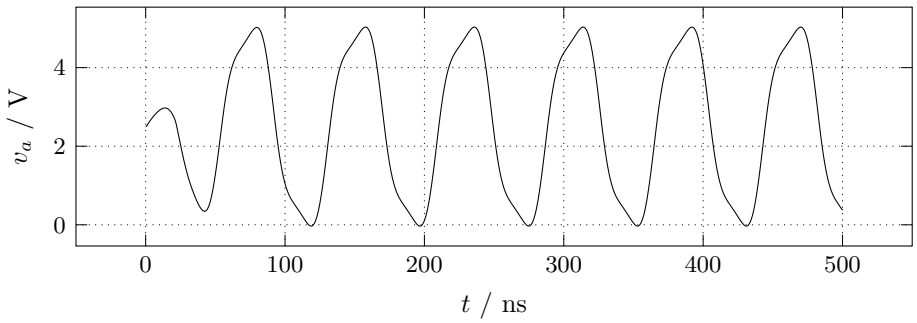



**LC tank.** Undriven, started from an inductor initial condition with `uic`. The trapezoidal rule is undamped, so the amplitude holds at $i_0\sqrt{L/C} = 31.6\,\text{mV}$ (dashed):

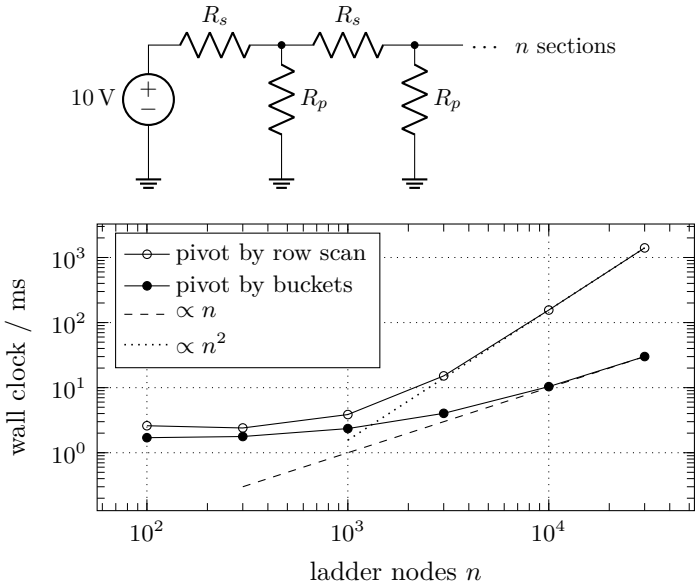



**Ring oscillator.** Three inverters in a loop; the gate capacitances set the stage delay. A short current kick moves the circuit off its metastable operating point, then the ring settles to a 78 ns period:






**Benchmark.** Operating point of resistor ladders of increasing size, wall clock of the whole process (parse, solve, print one column), best of several runs on an Apple M3, release build:




Below about $10^3$ nodes the process startup dominates both curves. The open circles are the first version of the solver, which scanned every remaining row per column to find the pivot and follows the $n^2$ guide; this plot is what exposed it. The bucket version of section 7 follows the $n$ guide through the last point, 30 ms for 30000 nodes, 47 times faster than the scan at that size.

## 13 Validation

`tests/cli.rs` 330 lines, 0 of 1000: tests are free

The test suite is 23 integration tests that run the built binary on real netlists and parse its CSV. Tests do not count toward the budget, so robustness lives there, and the suite prefers references that are independent of the implementation: closed-form solutions, physical invariants, and reductions computed separately in the test itself.

| test | reference |
|---|---|
| `op_divider` | voltage division, source current sign convention |
| `op_diode` | KCL closure against the Shockley equation |
| `op_mos` | saturation current $\frac{1}{2}K_P v_{ov}^2$, exact bias |
| `op_jfet` | JFET drain current $\beta V_{TO}^2$ at $v_{GS} = 0$ |
| `op_model_card` | $n$=2 diode drop, fails if the model is not resolved |
| `op_bjt` | $I_C = \beta_F I_B$ and exact npn/pnp mirror symmetry |
| `op_ladder` | 201 equal resistors, interior node voltage and current |
| `op_random_ladder` | LCG-randomized ladder vs. Thevenin reduction in the test |
| `op_latch` | cross-coupled NMOS latch, hard operating point |
| `dc_sweep` | linearity of the swept divider |
| `tran_rc`, `tran_rl` | step responses vs. $1 - e^{-t/\tau}$ |
| `tran_lte_fast_tau` | $\tau =$ `tstep`/10, LTE control must resolve the edge |
| `tran_uic_lc` | LC amplitude $i_0\sqrt{L/C}$ from initial conditions |
| `tran_pwl` | pwl triangle tracked exactly at every output time |
| `tran_diode_cap` | reverse diode `cjo` with $m$=0 as a linear RC |
| `ac_depletion_cap` | graded junction: $C = C_{j0}/2$ at $v = -3\,\mathrm{V}$ |
| `tran_ring_oscillator` | 3-stage CMOS ring with a stable period |
| `tran_lc_energy` | no numerical damping over twenty periods |
| `ac_rc` | $1/\sqrt{2}$ magnitude and $-45°$ at the corner |
| `print_selection` | column filtering |
| `clean_errors` | seven malformed decks: named errors, no panics |
| `loc_budget` | the budget itself |

Two tests carry the most weight. The randomized ladder builds a 30-stage resistor network from a seeded LCG, computes every node voltage by backward Thevenin reduction and forward division inside the test, and compares node by node at $10^{-6}$ relative: this validates the sparse factorization, the pivoting, and the fill-in handling on unstructured values with an algorithm that shares no code with the solver. The energy test runs the LC tank for twenty periods and rejects any numerical damping, which pins the choice of the trapezoidal rule: backward Euler fails this test by design.

## 14 The cut list

| feature | why it lost | est. lines |
|---|---|---|
| `.subckt` hierarchy | lost to the transistor models | ∼100 |
| Markowitz ordering | netlist order suffices at this scale | ∼80 |
| Gummel-Poon BJT | Ebers-Moll covers the teaching core | ∼80 |
| noise analysis | needs the adjoint system | ∼80 |
| `.param` expressions | a calculator is its own project | ∼60 |
| temperature sweeps | $V_T$ is a constant, by choice | ∼25 |
| current-based convergence check | limiting denial covers the failure mode | ∼15 |

Table 2: Features that did not fit, with the reason and the estimated cost.

The most consequential remaining cut is `.subckt`: real decks are hierarchical, and flattening them by hand is the main friction in practice. It would be the first addition if the budget grew.

On language: the device enum with exhaustive matching carries the parser, the assembly, and the output selection, and is the main reason the budget holds. C++ would trade the roughly 60 lines the `Num` trait and complex type cost against visitor or inheritance boilerplate around `std::variant`, roughly a wash with fewer static guarantees. Fortran is strong on dense

arrays, which this simulator does not use, and weak on the parser and the sparse row structure. Python would roughly halve the line count at a different performance class.

## 15 Closing

The final count is 1000 of 1000 lines, with the classic algorithm set complete: MNA, sparse LU with partial pivoting, two Verilog-A style contribution primitives carrying all device stamps, Newton with pnjlim and both continuation fallbacks as data-driven homotopy paths, trapezoidal integration with LTE control, breakpoints and damped restart steps, charge and flux based reactive state with graded junction capacitances, small-signal AC, eleven device letters with three source waveforms, four analyses, model cards, initial conditions, and output selection. The budget is spent exactly.